Read-in action in mixed Hermite space

Two-layer erf nets with a nondiagonal He1–He3 block J

Notes for discussion (notation as in Lipshitz MSc thesis)

October 20, 2026

Abstract

These notes specialize the thesis notation to a *two-layer* erf network and keep the teacher-aligned Hermite block of the first-layer kernel *nondiagonal*. The thesis mean-field treatment often works in a basis where that block is diagonal; here we keep the off-diagonal matrix element J_{13} explicit. We (i) write the posterior in the mixed He1–He3 mode space, (ii) integrate the readout to obtain a quadratic amplitude matrix G acting on the Hermite coefficients $c(w)$, (iii) obtain the scalar read-in action $V(w)$ along the teacher axis, and (iv) perform a Ginzburg–Landau expansion that identifies which invariant order parameters control the inner (center) versus outer well transitions, and how ε enters. The first-layer kernel J is the per-neuron activation kernel of the thesis (eq. (3.7) / (D.1)). The cavity resolvent is $\big(\sigma_A^2 J + \frac{\kappa}{P} I\big)^{-1}$.

Contents

1 Setup and notation **1**

2 Posterior in mixed (He1–He3) mode space **3**

3 Amplitude matrix G **3**

4 Hermite coefficients and the scalar read-in action **4**

5 Ginzburg–Landau treatment after stripping d **5**

5.1 Small-w expansion (inner / center transition) 5

5.2 Outer transition 6

5.3 Protocol to map the phase diagram 6

6 Summary **6**

1 Setup and notation

We work with a two-layer network with erf activations,

$$f_\Theta(x) = \sum_{i=1}^{N} a_i \, \Phi\big(w_i \cdot x\big), \qquad \Phi = \mathrm{erf}, \tag{1}$$

with read-in rows $w_i \in \mathbb{R}^d$ collected in $W \in \mathbb{R}^{N\times d}$ and readout $a \in \mathbb{R}^N$. (The three-layer objects B, h_k, ν_k of the thesis are absent.)

Data and target. Inputs $x \sim \mathcal{N}(0, I_d)$. The target lives in the span of the first two odd probabilists' Hermite polynomials along a unit teacher direction v_* ($\|v_*\| = 1$):

$$y(x) = \mathrm{He}_1(v_* \cdot x) + \varepsilon\, \mathrm{He}_3(v_* \cdot x), \tag{2}$$

with $\mathrm{He}_1(z) = z$ and $\mathrm{He}_3(z) = \frac{1}{\sqrt{6}}(z^3 - 3z)$ as in the thesis. We write Hermite feature maps $\varphi_1(x) = \mathrm{He}_1(v_* \cdot x)$ and $\varphi_3(x) = \mathrm{He}_3(v_* \cdot x)$.

Inner products. Dataset / population averages use the input measure $d\mu_x$:

$$\langle h \mid g \rangle_x := \int d\mu_x\, h(x)\, g(x), \qquad \|h\|_x^2 := \langle h \mid h \rangle_x. \tag{3}$$

Operator kernels act as $\big(Ag\big)(x) = \int d\mu_{x'}\, A(x, x') g(x')$.

Hyperparameters.

- P: number of training samples;
- κ: noise scale in the data term (Langevin temperature $T \equiv 2\kappa$);
- χ: train / loss scale in the data term;
- σ_A^2: readout prior variance per neuron;
- σ_w^2: prior variance of each read-in weight component (usually $\sigma_w^2 = 1$), so that the read-in prior term is $\frac{d}{2\sigma_w^2}\|W\|^2$.

Bayesian action.

$$S[a, W] = \frac{1}{2\sigma_A^2}\,\|a\|^2 + \frac{d}{2\sigma_w^2}\,\|W\|^2 + \frac{P\chi}{2\kappa}\,\big\|f - y\big\|_x^2. \tag{4}$$

The posterior is $p(a, W) = Z^{-1} e^{-S[a,W]}$ with $Z = \int da\, dW\, e^{-S}$. The Gaussian readout measure implied by the first term is $a \sim \mathcal{N}\big(0, \sigma_A^2 I\big)$.

First-layer kernel J. As in the thesis (eq. (3.7), and the arcsin evaluation (D.1)), J is the expectation of a single read-in neuron's activations:

$$J(x, x') := \mathbb{E}_{w \sim p(w|D)}\big[\Phi(w \cdot x)\, \Phi(w \cdot x')\big]. \tag{5}$$

There is no $1/N$ in this definition. In the large-d Hermite basis aligned with v_*, it is enough for the teacher (2) to retain the 2×2 block of Rayleigh quotients

$$J_{mn} := \langle \varphi_m \mid J \mid \varphi_n \rangle_x, \qquad m, n \in \{1, 3\}, \tag{6}$$

i.e. the finite matrix

$$J = \begin{pmatrix} J_{11} & J_{13} \\ J_{13} & J_{33} \end{pmatrix}. \tag{7}$$

The thesis often assumes a basis in which $J_{13} = 0$. We do *not*: mixed space means J_{13} is retained.

Remark 1 (Perp modes)**.** *Modes orthogonal to v_* remain (to leading large-d order) unadapted and are absorbed into an effective floor*

$$T = 1 + 2\,\frac{d-1}{d}\,\sigma_w^2 \tag{8}$$

in the formulae for c_1 and c_3 below; they do not enlarge the 2×2 teacher block.

2 Posterior in mixed (He1–He3) mode space

Project the network output and the target onto $\mathrm{span}\{\varphi_1, \varphi_3\}$. Write mode coefficients

$$f_m := \langle \varphi_m \mid f \rangle_x, \qquad y_m := \langle \varphi_m \mid y \rangle_x, \tag{9}$$

so that, for the target (2),

$$y = \begin{pmatrix} y_1 \\ y_3 \end{pmatrix} = \begin{pmatrix} 1 \\ \varepsilon \end{pmatrix} \tag{10}$$

(up to the conventional normalization of He_3; absolute scales can be absorbed into ε).

Restricted to this subspace, $f = (f_1, f_3)^\top$. Integrating the Gaussian readout $a \sim \mathcal{N}(0, \sigma_A^2 I)$ produces the mode-space action

$$S_{\mathrm{mix}}[f] = \frac{1}{2}\, f^\top \big(\sigma_A^2 J\big)^{-1} f + \frac{P\chi}{2\kappa}\, \|f - y\|^2, \tag{11}$$

with $\|f - y\|^2 = (f - y)^\top (f - y)$.

Cavity field. The equivalent-kernel mean is $\bar{f} = \sigma_A^2 J\big(\sigma_A^2 J + \frac{\kappa}{P} I\big)^{-1} y$. Define the cavity field by $\bar{f} = \sigma_A^2 J\, v$, so

$$\boxed{v := \Big(\sigma_A^2\, J + \frac{\kappa}{P}\, I_2\Big)^{-1} y} = \Big(\sigma_A^2\, J + \frac{\kappa}{P}\, I_2\Big)^{-1} \begin{pmatrix} 1 \\ \varepsilon \end{pmatrix}. \tag{12}$$

Write $R := \sigma_A^2\, J + \frac{\kappa}{P} I$, so $R_{mn} = \sigma_A^2\, J_{mn} + \frac{\kappa}{P}\delta_{mn}$, and $\Delta := \det R$. Then

$$v_1 = \frac{1}{\Delta}\big(R_{33} - R_{13}\, \varepsilon\big), \tag{13}$$

$$v_3 = \frac{1}{\Delta}\big(-\, R_{13} + R_{11}\, \varepsilon\big). \tag{14}$$

Both components depend on ε and on the mixing element J_{13}. If $J_{13} = 0$, then $R_{13} = 0$ and

$$v_1 = R_{11}^{-1} = \big(\sigma_A^2\, J_{11} + \tfrac{\kappa}{P}\big)^{-1}, \qquad v_3 = \varepsilon\, R_{33}^{-1} = \varepsilon\big(\sigma_A^2\, J_{33} + \tfrac{\kappa}{P}\big)^{-1}. \tag{15}$$

Learnability. The mode-space posterior mean is

$$\bar{f} = \sigma_A^2\, J\, v = \sigma_A^2\, J\Big(\sigma_A^2\, J + \tfrac{\kappa}{P} I\Big)^{-1} y. \tag{16}$$

In the eigenbasis of J with eigenvalues λ_m,

$$\tilde{L}_m = \frac{\sigma_A^2\, \lambda_m}{\sigma_A^2\, \lambda_m + \kappa/P}. \tag{17}$$

Learnability is controlled by the spectrum of J versus the ridge κ/P (with σ_A^2 weighting J), not by which wells of the eventual read-in potential are occupied.

3 Amplitude matrix G

With the Onsager piece omitted and $\delta = 1$, the He1–He3 amplitude of `laplace_amplitude_matrix` is

$$\boxed{G = G_{\mathrm{disc}} = \frac{\sigma_A^2}{N\chi}\big(-\chi^2\, vv^\top\big) = -\frac{\sigma_A^2 \chi}{N}\, vv^\top}, \tag{18}$$

with v from (12). A single neuron with coefficients c contributes $\frac{1}{2}\, c^\top G\, c = -\frac{\sigma_A^2 \chi}{2N}\, (v^\top c)^2$.

Definition of ϕ. Recall $R := \sigma_A^2 J + \frac{\kappa}{P} I$, so that $v = R^{-1} y$ with $y = (1, \varepsilon)^\top$. For a feature vector $c = (c_1, c_3)^\top$, write $R^{-1} c$ in the same He1–He3 components and define

$$\phi_1(c) := \big(R^{-1} c\big)_1 = R_{11}^{-1}\, c_1 + R_{13}^{-1}\, c_3, \tag{19}$$

$$\phi_\varepsilon(c) := \big(R^{-1} c\big)_3 = R_{31}^{-1}\, c_1 + R_{33}^{-1}\, c_3. \tag{20}$$

(R is symmetric, so $R_{13}^{-1} = R_{31}^{-1}$.) Thus $\phi_1(c)$ is the He1 component of $R^{-1} c$ and $\phi_\varepsilon(c)$ is its He3 component. Because $v^\top c = y^\top R^{-1} c$,

$$v^\top c = \phi_1(c) + \varepsilon\, \phi_\varepsilon(c). \tag{21}$$

The discrepancy quadratic form is therefore the explicit scalar

$$\boxed{\tfrac{1}{2}\, c^\top G_{\text{disc}}\, c = -\frac{\sigma_A^2 \chi}{2N} \Big(\phi_1(c)^2 + 2\varepsilon\, \phi_1(c)\, \phi_\varepsilon(c) + \varepsilon^2\, \phi_\varepsilon(c)^2 \Big)}. \tag{22}$$

4 Hermite coefficients and the scalar read-in action

Reduce each read-in row to the teacher coordinate $w := w_i \cdot v_*$. The overlaps $\langle \Phi \circ w \mid \mathrm{He}_n \circ v_* \rangle_x$ define the coefficients $c(w) = (c_1(w), c_3(w))$:

$$c_1(w) = \sqrt{\frac{4}{\pi}}\, \frac{w}{(T + 2w^2)^{1/2}}, \tag{23}$$

$$c_3(w) = -\sqrt{\frac{8}{3\pi}}\, \frac{w^3}{(T + 2w^2)^{3/2}}, \tag{24}$$

normalized so that, in a Laplace / mixture ansatz for $p(w)$,

$$J \approx \mathbb{E}_{w \sim p(w)} \big[c(w) c(w)^\top \big]. \tag{25}$$

(We do not assume a sign for c_3 beyond the conventional overall phase in (24).)

The effective single-neuron read-in action is the prior wall plus (22) evaluated on $c(w)$:

$$\boxed{V(w) = \frac{d}{2\sigma_w^2}\, w^2 - \frac{\sigma_A^2 \chi}{2N} \Big(\phi_1\big(c(w)\big)^2 + 2\varepsilon\, \phi_1\big(c(w)\big)\, \phi_\varepsilon\big(c(w)\big) + \varepsilon^2\, \phi_\varepsilon\big(c(w)\big)^2 \Big)} \tag{26}$$

Self-consistency closes by rebuilding J from the wells of V: locate wells $\{\mu_k, \sigma_k\}$ of V, assign Laplace mixture weights

$$\pi_k = \frac{m_k\, \sigma_k\, e^{-V(\mu_k)}}{\sum_\ell m_\ell\, \sigma_\ell\, e^{-V(\mu_\ell)}} \qquad (m_k = 2 \text{ for a } \pm\mu \text{ pair},\ m_k = 1 \text{ if } \mu_k = 0), \tag{27}$$

and set $J = \sum_k \pi_k\, \mathbb{E}_{w \sim \mathcal{N}(\mu_k, \sigma_k^2)} [c(w) c(w)^\top]$.

Where ε enters. Hold J fixed, and therefore hold R fixed. The definitions (19)–(20) are components of $R^{-1} c$, and R does not contain ε, so the two numbers $A := \phi_1(c)$ and $B := \phi_\varepsilon(c)$ do not depend on ε. The quadratic in (22) is then

$$Q(\varepsilon) := A^2 + 2\varepsilon\, AB + \varepsilon^2 B^2, \qquad Q(-\varepsilon) = A^2 - 2\varepsilon\, AB + \varepsilon^2 B^2. \tag{28}$$

Subtracting gives

$$Q(\varepsilon) - Q(-\varepsilon) = 4\varepsilon\, AB = 4\varepsilon\, \phi_1(c)\, \phi_\varepsilon(c). \tag{29}$$

The cross term is the unique odd piece of Q because it is the only piece that changes sign, and A, B were not allowed to change. That proviso matters. Once J is rebuilt from the wells of V, both ϕ's depend on ε through R^{-1}, and (29) no longer isolates the sign flip.

5 Ginzburg–Landau treatment after stripping d

Define the d-stripped potential

$$\tilde{v}(w) := \frac{2\sigma_w^2}{d}\, V(w). \tag{30}$$

Phase structure of V coincides with that of $\tilde{v}$.

Definition of C. Substitute (26) into (30). The prefactor of $-\big(\phi_1(c) + \varepsilon\phi_\varepsilon(c)\big)^2$ is a pure combination of the external scales $(\sigma_w^2, \sigma_A^2, \chi, N, d)$; it does not depend on w, on J, or on ε. We therefore define

$$C := \frac{\sigma_w^2\, \sigma_A^2\, \chi}{Nd}, \tag{31}$$

so that the stripped action is

$$\tilde{v}(w) = w^2 - C\,\Big(\phi_1\big(c(w)\big) + \varepsilon\,\phi_\varepsilon\big(c(w)\big)\Big)^2. \tag{32}$$

C is the natural dimensionless coupling: it measures how strongly the readout-integrated data term (scaled by $\sigma_A^2\chi/N$) competes with the read-in prior wall w^2 after d has been stripped. The ridge that enters v is κ/P, with J weighted by σ_A^2 in $R = \sigma_A^2\, J + \frac{\kappa}{P} I$. An invariant ray holds $\big(C,\, \kappa/(\sigma_A^2 P)\big)$ fixed while dialling ε, or scans P at fixed $(\sigma_w^2, \sigma_A^2, \kappa, \chi, N, d)$.

5.1 Small-w expansion (inner / center transition)

Near $w = 0$,

$$c_1(w) = \alpha_1 w + O(w^3), \qquad c_3(w) = \alpha_3 w^3 + O(w^5), \tag{33}$$

with α_1, α_3 fixed by (23)–(24) and T. By (21) the combination in (32) has the same Taylor series as $v^\top c(w)$, namely

$$\big(\phi_1(c(w)) + \varepsilon\,\phi_\varepsilon(c(w))\big)^2 = (v_1\alpha_1)^2 w^2 + 2v_1v_3\alpha_1\alpha_3\, w^4 + (v_3\alpha_3)^2 w^6 + O(w^8). \tag{34}$$

Expanding (32) therefore gives

$$\tilde{v}(w) = a_2 w^2 + a_4 w^4 + a_6 w^6 + O(w^8), \tag{35}$$

with

$$a_2 = 1 - C\,(v_1\alpha_1)^2, \tag{36}$$

$$a_4 = -2C\, v_1 v_3 \alpha_1 \alpha_3, \tag{37}$$

$$a_6 = -C\,(v_3\alpha_3)^2. \tag{38}$$

Order parameters. The Landau coefficients depend on v only through the combinations Cv_1^2, Cv_1v_3, and Cv_3^2. We therefore introduce the dimensionless order parameters

$$\psi_1 := \sqrt{C}\, v_1, \qquad \psi_3 := \sqrt{C}\, v_3, \tag{39}$$

i.e. the resolvent components v_m measured in units of the coupling C defined in (31). Equivalently, $\psi_1^2 = Cv_1^2$, $\psi_3^2 = Cv_3^2$, $\psi_1\psi_3 = Cv_1v_3$, and (36)–(38) become

$$a_2 = 1 - (\alpha_1\psi_1)^2, \qquad a_4 = -2\alpha_1\alpha_3\,\psi_1\psi_3, \qquad a_6 = -(\alpha_3\psi_3)^2. \tag{40}$$

Thus (ψ_1, ψ_3) are the order parameters of the Ginzburg–Landau theory: they absorb all dependence on (v_1, v_3) and on the external scales packaged in C. Expanding v_1v_3 from (13)–(14) gives

$$v_1v_3 = \frac{1}{\Delta^2}\Big[-R_{33}R_{13} + \big(R_{11}R_{33} + R_{13}^2\big)\varepsilon - R_{11}R_{13}\,\varepsilon^2\Big], \tag{41}$$

which is the product of (13) and (14). At fixed J the entries of R do not depend on ε, so the term linear in ε is odd and the other two terms are even. The same split applies to v_1^2 and v_3^2. From (13)–(14), each of v_1 and v_3 is an affine function of ε as soon as $R_{13} \neq 0$, and therefore $a_2 \propto v_1^2$ and $a_6 \propto v_3^2$ each acquire an odd piece. Only the diagonal case $J_{13} = 0$ (hence $R_{13} = 0$) makes v_1 independent of ε and v_3 proportional to ε, in which case a_4 is the only odd Landau coefficient.

Landau diagnostics.

Phenomenon	Condition on (a_2, a_4, a_6)	Controlling order parameters
center spinodal	$a_2 = 0$, $a_4 > 0$	ψ_1^2
inner / central wells	$a_2 > 0$, $a_4 < 0$, $a_6 > 0$	$\psi_1\psi_3$ (He1–ε cross)
outer wells	large-$\lvert w\rvert$ / finite-μ analysis	$\lvert\psi_3\rvert$, J_{33}, κ/P, σ_A^2
learnability	$\tilde{L}_m = \dfrac{\sigma_A^2\lambda_m}{\sigma_A^2\lambda_m + \kappa/P}$	spectrum of J vs κ/P

With $\varepsilon > 0$ and lab mixing, a_4 (fed by $\psi_1\psi_3$) leaves the **inner/center** reorganization live, while the cubic channel $|\psi_3|$ opens **outer** minima at large $|w|$—two landscape transitions. Changing the sign of ε at fixed J changes every Landau coefficient that has an odd piece: a_4 in general, and also a_2 and a_6 when $J_{13} \neq 0$. The sign flip is isolated in the inner transition only in the diagonal block.

5.2 Outer transition

The small-w series cannot see outer wells. One either matches the large-$|w|$ asymptotics of $\tilde{v}(w)$, dominated by $-C(v_3 c_3(w))^2$, or introduces a finite-well order parameter $m = \mu^2$ and expands an effective potential $U(m)$. Criticality is a condition on $\big(|\psi_3|,\, J_{33},\, \kappa/P,\, \sigma_A^2\big)$, not on a_2.

5.3 Protocol to map the phase diagram

1. Fix an invariant ray: strip d; hold C and $\kappa/(\sigma_A^2 P)$.
2. Solve self-consistent $J \Rightarrow v = (\sigma_A^2\, J + \frac{\kappa}{P} I)^{-1} y \Rightarrow (\psi_1, \psi_3) = (\sqrt{C}\, v_1, \sqrt{C}\, v_3)$.
3. Compute (a_2, a_4, a_6) from the small-w expansion of $\tilde{v}$.
4. Draw boundaries in the order-parameter plane $(\psi_1^2, \psi_1\psi_3)$ or (a_2, a_4).
5. Overlay the outer boundary from a large-w / finite-μ condition.
6. Scan sign(ε) at fixed $|\varepsilon|$ and at fixed J: the odd piece of Q is the cross term (29). In the Landau coefficients the same sign change moves a_4, and also a_2 and a_6 when $J_{13} \neq 0$.

6 Summary

- Mixed-space posterior: keep the full 2×2 block J including J_{13}.
- Cavity field: $v = \left(\sigma_A^2\, J + \dfrac{\kappa}{P} I\right)^{-1} \begin{pmatrix} 1 \\ \varepsilon \end{pmatrix}$.
- Amplitude: $G = G_{\text{disc}} = \dfrac{\sigma_A^2}{N\chi}(-\chi^2 v v^\top) = -\dfrac{\sigma_A^2 \chi}{N}\, v v^\top$.
- For $c = (c_1, c_3)^\top$ and $R = \sigma_A^2\, J + \frac{\kappa}{P} I$: $\phi_1(c) = (R^{-1}c)_1$ and $\phi_\varepsilon(c) = (R^{-1}c)_3$.

- Discrepancy: $\frac{1}{2}c^\top G_{\mathrm{disc}} c = -\dfrac{\sigma_A^2 \chi}{2N}\big(\phi_1^2 + 2\varepsilon\,\phi_1\phi_\varepsilon + \varepsilon^2\phi_\varepsilon^2\big)$. At fixed J, $Q(\varepsilon) - Q(-\varepsilon) = 4\varepsilon\,\phi_1\phi_\varepsilon$.
- Read-in action: prior wall plus that discrepancy on $c(w)$.
- Coupling $C = \sigma_w^2\sigma_A^2\chi/(Nd)$ is the prefactor of $-(\phi_1 + \varepsilon\phi_\varepsilon)^2$ in $\tilde{v}$; order parameters $\psi_m = \sqrt{C}\,v_m$ make (a_2, a_4, a_6) depend on (ψ_1, ψ_3) alone (inner $\leftrightarrow$ a_4 $\leftrightarrow$ $\psi_1\psi_3$; outer $\leftrightarrow$ $|\psi_3|$; learning $\leftrightarrow$ spectrum of J versus κ/P).

These notes are intended for discussion with the thesis advisor; they extend the diagonal-J treatment of the MSc thesis to mixed He1–He3 space for the two-layer read-in problem.